

Rajendra Maharjan

COMPUTATIONAL CHEMISTRY · VISITING ASSISTANT PROFESSOR

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Summary

I am currently a Visiting Assistant Professor at Grinnell College, where I teach Physical Chemistry and General Chemistry. I received my PhD in Computational Chemistry from the Department of Chemistry at Oklahoma State University in Fall 2023. My research focuses on the molecular level organization of water in pure water and aqueous solution in presence of hydrophobic particle. Additionally, I investigate the dipole moment distribution of water in the condensed phase in response to various environmental factors.

Education

Oklahoma State University

PHD IN COMPUTATIONAL CHEMISTRY

Stillwater, Oklahoma

Aug, 2017 - Dec, 2023

- Dissertation: 'Investigation of Hidden Structure and the Dipole Moment Distribution in Liquid Water and Ice'

Tribhuvan University

MASTER OF SCIENCE (MS) IN PHYSICAL CHEMISTRY

Kathmandu, Nepal

January 2011 - May, 2014

- Thesis: 'Preparation and Characterization of Solid-State Electrochemical Sensor for the Determination of Ni(II)

Tri-Chandra Multiple Campus

BACHELOR OF SCIENCE (BS) MAJOR IN CHEMISTRY AND MICROBIOLOGY

Kathmandu, Nepal

August, 2006 - March, 2010

Research Experience

Oklahoma State University

GRADUATE TEACHING/RESEARCH ASSISTANT

Stillwater, OK

Aug, 2017 - Dec, 2023

Study of Water in Liquid, Supercooled, and Crystalline Water

Project 1: Assessing Order in Liquid, Supercooled, and Crystalline Water

- Developed an algorithm (nucleation_tracker written in both C++ and Python) to study network topology analysis and water structural ordering using an order parameter.
- Monitor closed rings distribution in liquid and supercooled water using different empirical potential water models at different temperatures.
- Studied intrinsic hydrogen bonded polygons in different known ice polymorphs and render the Pov-Ray image unique to each ice polymorphs.
- Studied tetrahedral order parameter for water ordering in liquid, supercooled liquid, and ice.

Project 2: Estimation of Dipole Moment Distribution of Water in Condensed Phase

- Determine the dipole moment distribution of water in liquid and ice using quantum approach (QM) and ab-initio molecular dynamics (AIMD).
- Studied the polarization effect from the environment on enhancing the dipole moment of gas phase water in condensed system.
- Developed an algorithm based on tabulated quantum calculation to determine the dipole moment of water at different temperatures that is comparable to pure QM calculations.
- Employed ice Ih and ice VII as a toy model to study moment distribution dependent on density, quality of hydrogen bonds, and hydrogen bond distance.

Project 3: Crystal Seed Assisted Ice Nucleation in Supercooled Water

- Stochastic growth of ice using crystal seed and direct coexistence method from pure water.
- Network topology analysis and water structural ordering during molecular dynamics (MD) simulation.
- Study of crystal defects during ice formation.

Project 4: Range of Water Ordering in Response to the Solvated Hydrophobic Particle

- Studied how the hydrophobic particle enhance the water ordering using 'nucleation_tracker' program
- Carefully monitor the constructive interference behavior of two hydrophobic particles on water ordering.
- Studied chaotropic effect of ions on water ordering.

Masters Thesis**Project:** Preparation and Characterization of Solid-State Electrochemical Sensor for the Determination of Ni(II)

- Development of electrochemical sensor and improve its detection limit.
- Used alkaline medium to prepare the nano-disc sensor and tested its performance in different solutions.
- Used potentiometer to monitor the response of the sensor to the solution.

Teaching Experience

Grinnell College

1115 8th Avenue, Grinnell, IA

VISITING ASSISTANT PROFESSOR

Aug. 2024 - Present

Job Description:

- Taught Physical Chemistry (Chm 363) and General Chemistry (Chm 129) with labs, an undergraduate course averaging 36 students per semester.
- Developed quizzes, exams, and homework.
- Revised the syllabus to meet accreditation standards.
- Conducted and supervised laboratory.
- Carried out research with undergraduate students.
 - * Range of Water Ordering in the Presence of Hydrophobic Particle.
 - * Dipole Moment Distribution of Water in Condensed Phase.
 - * Investigation of Phase Transition in Supercooled Water and Ice Nucleation in Supercooled Water.

Oklahoma State University

Stillwater, OK

TEACHING ASSISTANT

Aug. 2017 - Dec. 2023

Labs: Chem 1314 and Chem 1515

Lecture Class: Physical Chemistry-1

- Conducted laboratory experiments
- Provided grades and comments on student's assignments, lab reports and exams.
- Held office hours to review class materials and discuss students' questions.

Higher Secondary School and Colleges

Kathmandu, Nepal

LECTURER, CHEMISTRY

Jan. 2014 - Jun. 2017

Colleges Taught:

- Jaya Multiple Campus, Gokarneshwor, Kathmandu, Nepal. (Duration: April, 2014 - April, 2016)
- Janasewa Higher Secondary School, Kirtipur, Kathmandu, Nepal. (Duration: June 2016 - June, 2017)
- Classic Academy Higher Secondary School, Chabahil, Kathmandu, Nepal. (Part time)
- Jubilant College and Research Center, Kalimati, Kathmandu, Nepal. (Part time)
- Shankarapur Hospital and Academy, (CTEVT program) Jorpati, Kathmandu, Nepal (Part time)

Job Descriptions:

- * Physical Chemistry and General Chemistry with labs, an undergraduate course averaging 30 students per year.
- * Developed quizzes, exams, and homework.
- * Revised the syllabus to meet accreditation standards.
- * Conducted and supervised laboratory.

Undergraduate Students Advised

MENTOR

- Can Yang, *Range of Water Ordering in Hydrophobic Solvation of Inert Gases*
Role: Advisor
Research Proposal Submitted: August, 2025
Poster Presented in MWRM-2026
https://www.mwrm2025.org/_files/ugd/5ef6d9_bc377506186f4e1fa91a14d259bed12b.pdf
- Sarah Fory, *Ice Crystal Growth in Solutions*
Role: Co-Advisor
Research Proposal Submitted: April, 2023
Poster Presented in 2023 Annual Undergraduate Research Symposium, Conoco-Phillips Alumni Center, Oklahoma State University

Professional Experience

TechnoMark Solutions Pvt. Ltd.

RESEARCH AND DEVELOPMENT CHEMIST

Kapan, Kathmandu, Nepal

Feb, 2012 - Dec, 2012

Manager's Name: Naresh Sharma

Hours worked per week: 10 hours

Job Description:

- Conducted the project on 'Chlor-alkali PEM fuel cell'
- Prepared and fabricated a fuel-cell model and tested its efficiency and durability in alkaline medium.

Skills

- **Programming Skills:** Python Programming for numerical calculations, Basic C++ programming, Basic Matlab programming, R-programming, Linux/Unix Programming
- **Software Carpentry Certified Instructor:** Instructor and helper for software carpentry workshops for 'Unix Shell', 'Python Programming', and 'Version Control'
- Statistical data analysis and data visualization in Python, R, and Advanced MS Excel
- **Molecular Dynamics Simulation:** Used GROMACS and OpenMD packages for MD simulations
- **Quantum Mechanics:** Used 'Gaussian/g16' and NWChem for pure QM calculations and 'CP2K' for ab-initio molecular dynamics (AIMD)
- **Computational tools:** Used Chimera, Jmol, and Avogadro for molecule visualization and modeling, Inkscape for creating vector images, XM-Grace, Packmol
- Typesetting/ Document Preparation/Referencing: LaTeX, MS Office, EndNote
- **Others:** GNU Image Manipulation Program (GIMP) and Adobe photoshop for image processing, Version Control: Git, GitHub

Seminar

Chem-6011: Hidden Structures and the Dipole Moment Distribution in Liquid Water and Ice. Department of Chemistry Seminar, May 23, 2023.

Chem-5011: Potentiometric Sensor Development for quantification of Cd(II) and other Heavy Metal Ions. Department of Chemistry Seminar, March 26, 2019.

Manuscripts & Publication

Maharjan R; Williamson C; Fennell C.J. Assessing Order in Liquid, Supercooled, and Crystalline Water. The Journal of Physical Chemistry B. 2026.

Rajendra Maharjan and Christopher J. Fennell. Dipole Moment Distribution Analysis in Liquid Water and Ice. Manuscript in Preparation.

Rajendra Maharjan, Sarah Foy, Ibrahim Asad, and Christopher J. Fennell. Seeding Ice Crystallization in Molecular Simulations. Manuscript in Preparation.

Maharjan, R.; Nakarmi, K. B.; and Yadav, A. P. Preparation and Characterization of Solid-State Electrochemical Sensor for the Determination of Ni(II). *J. Nepal Chem. Soc.* **2016**, 35, 104–112. <https://doi.org/10.5281/zenodo.15794577>.

Conference Presentations

Can Yang and **Rajendra Maharjan**. Range of Water Ordering in Hydrophobic Solvation of Inert Gases. MAP Research Poster Presentation, Grinnell College, 31 July, 2025. **Poster**

Rajendra Maharjan. Hidden Structure and the Dipole Moment Distribution in Liquid Water and Ice. On-Campus Interview Presentation. Grinnell College, Grinnell, IA, 3 April, 2024. **Oral**

Sarah Foy, **Rajendra Maharjan**, and Christopher J. Fennell. Ice Crystal Growth in Solutions. Annual Undergraduate Research Symposium, Conoco-Phillips Alumni Center, Oklahoma State University, April, 2023. **Poster**

Rajendra Maharjan and Christopher J. Fennell. Dipole Moment Distribution Analysis in Liquid Water and Ice. ACS Southwest Regional Meeting (SWRM), Baton Rouge, LA, November, 2022. **Poster**

Rajendra Maharjan and Christopher J. Fennell. Order in Water and its Range in Hydrophobic Solvation. ACS Midwest Regional Meeting (MWRM), Springfield, MO, October, 2021. **Poster**

Rajendra Maharjan and Christopher J. Fennell. Algorithmic Investigation of Order in Liquid, Crystalline, and Supercooled Water. Molecular Modeling and Materials Design (M3D) Conference, Department of Chemical Engineering, Oklahoma State University, July, 2020. **Oral**

Rajendra Maharjan and Christopher J. Fennell. Identification and Quantification of Water Ordering in Liquid, Crystalline, and Supercooled Water. The 2019 Annual Southwest Theoretical and Computational Chemistry Meeting, Norman, OK, September, 2019. **Poster**

Rajendra Maharjan and Christopher J. Fennell. Mapping Order in Liquid and Crystalline Water. ACS Midwest Regional Meeting (MWRM), Wichita, Kansas, October, 2019. **Poster**

Rajendra Maharjan, Casey Williamson, and Christopher J. Fennell. Seeding Ice Crystallization in Molecular Simulations. Department of Chemistry, Oklahoma State University, May, 2019. **Poster**

Rajendra Maharjan, Casey Williamson, and Christopher J. Fennell. Seeding Ice Crystallization in Molecular Simulations. 64th Pentasectional Meeting of the Oklahoma Sections of the ACS (Norman, OK), February, 2019. **Poster**

Rajan Timilsina, Krishna Badan Nakarmi, **Rajendra Maharjan**, and Amar Prasad Yadav. Effect of Back Contact on the response of Ni(II) Ion-Selective Electrodes. NAST - Nepal Academy of Science and Technology (7th National Conference of Science and Technology), O-PY-1-821, 2016. **Oral**

Rajendra Maharjan, Amar P. Yadav. Preparation and Characterization of Solid-State Electrochemical Sensor for the Determination of Ni(II). Advanced Materials and Nanotechnology for Sustainable Development, Nepal Chemical Society and Tribhuvan University, Kathmandu, Nepal, November, 2014. **Oral**

Workshops Attended

i-CoMSE MD/MC Summer School

*Oklahoma State University,
Stillwater, OK*

HANDS-ON FUNDAMENTAL CONCEPTS IN MOLECULAR DYNAMICS AND MONTE-CARLO SIMULATIONS

July 10 - 15, 2022

This Summer workshop focused on the theoretical principles of Monte Carlo (MC) and Molecular Dynamics (MD) simulations, along with techniques for computing macroscopic properties such as self-diffusion coefficients, viscosity, ionic conductivity, and vapor-liquid equilibria. We used Gromacs for MD simulations and 'Cassandra' for MC simulations during the hands-on tutorial sessions. I learned the theoretical foundations and practical methods of MD and MC techniques, as well as approaches for executing those software packages.

Telluride School on Theoretical Chemistry (TSTC)

Virtual format

SUMMER PROGRAM ON MOLECULAR DYNAMICS, ELECTRONIC STRUCTURES, STATISTICAL MECHANICS, AND BIOPHYSICS

July 19-23, 2021

This five-day summer workshop featured approximately six to eight hours of daily lectures on electronic structures, statistical mechanics, molecular dynamics, and biophysics. It was targeted students interested in pursuing careers in theoretical chemistry during their PhD programs and Postdoctoral positions. The program included hands-on problem solving sessions with examples solved during each session, additionally provided with numerous practice problems after the sessions.

The Carpentries Instructor Training

Stillwater, OK

HANDS-ON TRAINING IN R, PYTHON, THE UNIX BASH SHELL, SQL, OPENREFINE, SPREADSHEET SOFTWARE, AND GIT

July 18-19, 2019

This training prepared participants to become certified Carpentries instructors. The workshop introduced evidence-based best-practices in teaching, helping us create a positive learning environments, practice and develop our teaching skills, integrate into the Carpentries community, and apply skills in Carpentries workshops. This was training for teaching, not technical training. Participants must be familiar with at least one of the technologies incorporated in the Carpentry workshop (R, Python, the Unix bash shell, SQL, OpenRefine, spreadsheet software, and/or Git) before taking the Instructor Training.

Honors & Awards

OKLAHOMA STATE UNIVERSITY

- Graduate Abstract Writing - Honorable Mention**, Recognized with honorable mention for excellence in the graduate-level abstract writing contest, organized annually by the Graduate College of Oklahoma State University across the campus. In partnership with the OSU Writing Center, the Graduate College hosts the abstract writing contests. Students who attend a workshop for either technical (disciplinary) abstracts or public abstracts are eligible to compete. This award included a \$100 prize.
- 2023 Stillwater, OK
- Graduate Technical Abstract Writing - 2nd position**, My abstract was awarded in the technical clarity section for writing on my ongoing PhD research topic. This annual competition is organized by the Graduate College of Oklahoma State University across the campus. In partnership with the OSU Writing Center, the Graduate College hosts the abstract writing contests in the fall and spring semesters. Students who attend a workshop for either technical (disciplinary) abstracts or public abstracts are eligible to compete. This award include a \$200 prize.
- 2021 Stillwater, OK
- Outstanding Poster Award - 2nd position**, The poster session was organized by the Department of Chemistry at OSU, where my poster received the 2nd position. This award included a \$1,000 prize.
- 2019 Stillwater, OK

TRIBHUVAN UNIVERSITY

- University Grants Commission (UGC Nepal)**, My Master's Thesis research has been funded by the University Grants Commission. My Thesis, entitled "Preparation and Characterization of Solid-State Electrochemical Sensor for the Determination of Ni (II)", was funded under the "Thesis Support Grants UGC - 2012" program with \$300.
- 2014 Bhaktapur, Nepal
- 2nd-year Academic Excellence Scholarship (Partial Tuition Waiver) for the M.Sc Chemistry**, The Central Department of Chemistry at Tribhuvan University annually awards a \$300 worth scholarship to the top 20 students (out of 90) who perform outstandingly in their second year MS program. I ranked 16th in my second-year MS program.
- 2012 Kathmandu, Nepal
- 1st-year Academic Excellence Scholarship (Partial Tuition Waiver) for the M.Sc Chemistry**, The Central Department of Chemistry at Tribhuvan University annually awards a \$300 worth scholarship to the top 20 students (out of 90) who perform outstandingly in their first-year MS program. I ranked 9th in my first-year MS program.
- 2010 Kathmandu, Nepal

Professional & Student Affiliations

American Chemical Society

REGULAR MEMBER

2020 - 2021

Nepal Chemical Society

LIFE MEMBER

Kathmandu, Nepal

Participated in local meetings and the International Conference on Advanced Materials and Nanotechnology for Sustainable Development (November 4–6, 2024, Kathmandu, Nepal), organized by the Nepal Chemical Society and Central Department of Chemistry.

Nepalese Student Association

WEBMASTER

Stillwater, OK

2020 - 2021

Served as Webmaster, designing program cards and invitation letters, managing emails, and helping organize events

References

Christopher J. Fennell, PhD

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